Does the Market Already Know? Forecast Encompassing in Live Baseball Betting Markets

This paper asks whether publicly observable baseball information predicts outcomes, or merely predicts what the market already knows.

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Working Paper · July 2026

Abstract

How completely does a high-frequency market capitalize public information? Live, in-play sports betting offers an unusually clean setting in which to ask, because information arrives as discrete, well-valued events, prices update continuously, and every contract reaches a known terminal payoff within hours. Yet the market efficiency literature remains concentrated on pregame prices, leaving the calibration of live markets and their incorporation of in-game information comparatively unstudied. We treat a sharp live betting line as an incumbent forecast and apply the Chong and Hendry (1986) forecast encompassing framework: a public variable earns its place only if it improves on the market's own forecast of remaining runs, not merely if it predicts the outcome. Using 163 Major League Baseball games from June 2026, with one-minute live total trajectories, pitch-level measurement, and full play-by-play, we test whether any publicly observable baseball state variable (times through the order, velocity decline, bullpen fatigue, line-drop reversion, alternate-line skew, early-run anchoring, weather, park) carries information the market has not already priced. None does. The market's forecast error is not predictable out of sample from any variable we measure (R^2 of −0.037; a Clark-West nested comparison does not favor the augmented model), and the design has power to rule out moderate incremental information. The one variable that appears to beat the market, a starter's velocity decline, is shown to be post-treatment selection bias: its apparent predictive power collapses to a coin flip once measured before the outcome selects the sample. An event-level transfer function shows the line absorbing each information shock by an approximately uniform magnitude, consistent with proportional capitalization. We name the resulting boundary the efficient frontier of public information, the point at which additional public variables stop improving on the market forecast, and we characterize it precisely at the one-minute, single-book resolution of our data. We release the Third Turn Protocol, a sequential validation ladder built from standard forecast-evaluation tools, and an accompanying benchmark dataset.

Keywords: market efficiency; high-frequency betting markets; forecast encompassing; incremental information; public-information capitalization; calibration; reproducible benchmark. *JEL codes:* C53 (forecasting), G14 (information and market efficiency), Z23 (sports economics).

1. Introduction

Live, in-play wagering has grown from a curiosity into a large and rising share of sportsbook handle; industry reporting places it at or near half of volume in mature markets. The academic literature has not kept pace. The economics of wagering markets, surveyed by Sauer (1998), is built mostly on pregame moneylines and point spreads, where a single closing price can be compared against a realized outcome. Live markets pose a different problem. Prices update continuously as the game state evolves, so the object of interest is not one forecast but a stream of them, and the natural questions concern calibration and the speed and completeness of information incorporation rather than a one-shot verdict. Whether live markets efficiently absorb the publicly observable information generated during a game, and how one would even test such a claim, remains largely open.

Baseball is an unusually clean laboratory for the question, and for much the same reasons that have long drawn financial economists to betting markets (Thaler and Ziemba, 1988; Sauer, 1998). Each contract reaches an unambiguous terminal payoff within hours, so there is a ground truth against which every forecast can be scored, without the discounting, horizon, and never-realized-fundamental problems that complicate the same test in equities. The game unfolds as a sequence of discrete events with well-established run values (the RE24 base-out run expectancy table and linear weights), so the informational content of each event can be quantified rather than estimated. Pitch-level measurement makes within-game state observable in fine detail: velocity, pitch count, times through the order. In this setting the efficiency question sharpens to a concrete one, how completely and how quickly a live price capitalizes the public information the game generates. The sport also carries a rich stock of public prior belief, most prominently the times-through-order penalty (Tango, Lichtman, and Dolphin, 2007), the widely held view that a starting pitcher degrades sharply the third time he faces a lineup. If the live market underprices that deterioration, the live Over is a profitable bet. This belief is our entry point: concrete, popular, and plausible.[1]

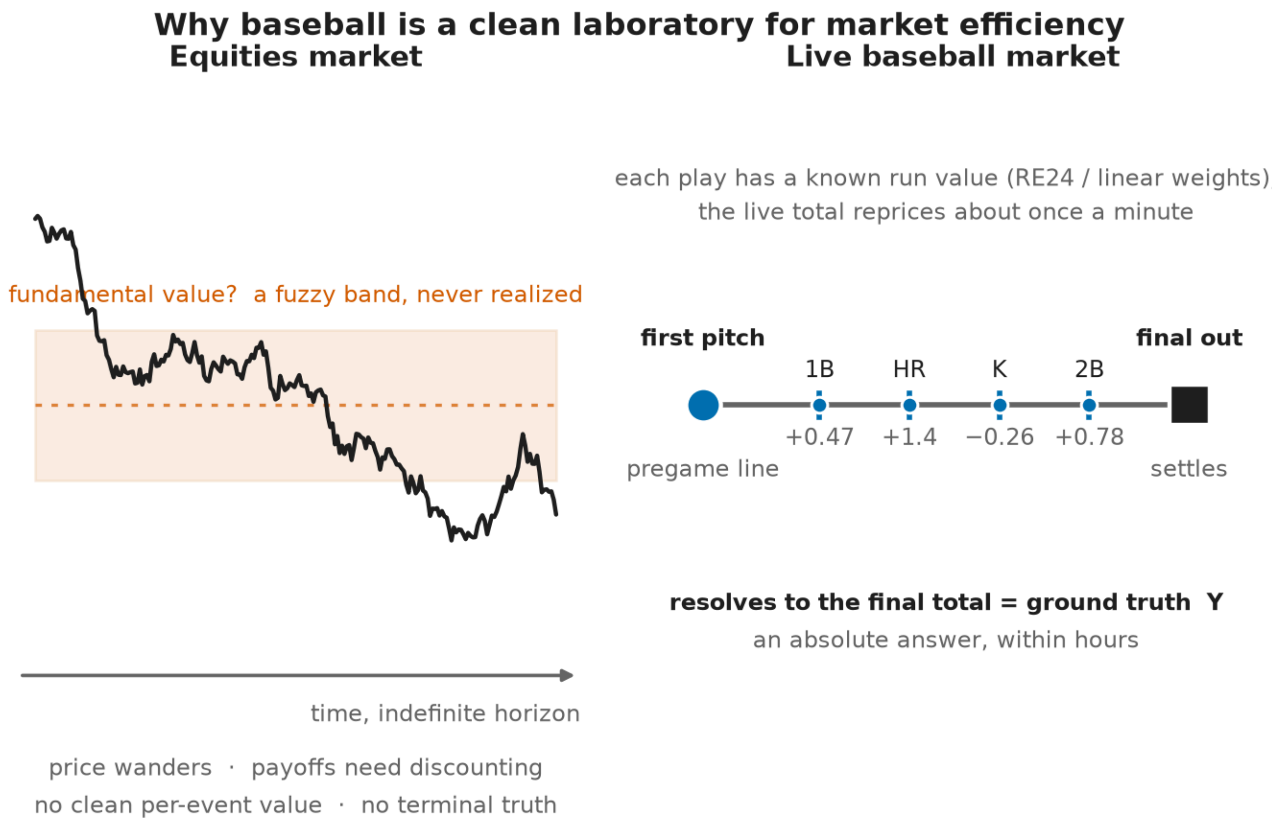

Figure 1. Why baseball is a clean laboratory for market efficiency. Left: an equity price wanders around a fundamental value that is never realized, and its payoffs must be discounted over an indefinite horizon. Right: a live baseball game is a sequence of events with known run values (RE24 and linear weights), the live total reprices about once a minute, and within hours the game settles at a final total that is an absolute ground truth, Y. The terminal payoff and the clean per-event valuation, largely absent in asset markets, are what let the efficiency question be posed as a clean forecast comparison.

The question that matters, however, is not whether such variables predict scoring. Many of them do. The question is whether they carry incremental predictive information once one conditions on the forecast already embedded in a sharp live market. The distinction between predicting an outcome and improving an existing forecast of that outcome is the conceptual center of this paper, and it is routinely blurred in applied betting research, where an in-sample association is often treated as evidence of an edge. We take the opposite stance. An in-sample signal is the weakest admissible evidence, and each candidate variable must survive progressively stronger tests, ending with a forecast encompassing test against the market itself. Within the limits of our data, no variable does. We read this not as a failure to find an edge but as a measurement of the boundary at which publicly observable baseball information stops adding value beyond the market. We map that boundary and bind it to the conditions under which it was measured: 163 games in a single month, at one-minute cadence, from a single sharp-book feed.

This paper makes three contributions. The first is empirical: using forecast encompassing to condition on the market's own forecast, we map the efficient frontier of public information for live baseball totals and find that it binds tightly. No publicly observable baseball variable we test (times through the order, velocity decline, bullpen fatigue, line-drop reversion, alternate-line skew, early-run anchoring, weather, park) improves on the sharp live forecast of remaining runs; the market's forecast error is not predictable from any of them out of sample. The most instructive case is velocity decline, the one variable that appears to beat the market until it is exposed as an artifact of post-treatment selection, a result we develop in detail because it shows precisely how raw prediction misleads. The second contribution is methodological: the individual tools are standard, and our contribution is their integration into a single escalating protocol, the Third Turn Protocol (signal, robustness, out of sample, debiasing, conditional testing, forecast encompassing, transfer function), which shifts the burden of proof from demonstrating prediction to demonstrating incremental information beyond an existing forecast. The protocol applies to any market with a sharp public forecast and observable state. The third is infrastructure: we release the cleaned data and feature schema as the Third Turn Benchmark Dataset (v1), with reference implementations that operationalize the protocol, so that future hypotheses can be evaluated against the same yardstick rather than rebuilt from scratch.

The remainder of the paper is organized around one question, which every section serves: do publicly observable baseball state variables contain incremental predictive information about remaining runs beyond the forecast embedded in a sharp live betting market?

Box 1. The Third Turn Protocol in brief

1. Does the candidate variable predict the outcome at all? If not, discard it.
2. Does the association survive robustness checks? If not, discard it.
3. Does it survive debiasing? If not, it is a selection artifact, not an effect.
4. Does it improve the market's own forecast? If yes, it is genuinely new information. If no, it is already in the price.

Only a variable that clears the final gate carries information the price does not already hold. Every hypothesis in this study reaches that gate and stops.

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2. Related Work

This section is deliberately brief. Its job is to establish a gap, not to survey three literatures, and we draw on each only for what the design requires.

Wagering-market efficiency. The broad finding of the wagering literature (Sauer, 1998) is that betting prices are accurate but not perfect: systematic deviations such as the favorite-longshot bias exist, yet rarely survive transaction costs. For baseball specifically, Woodland and Woodland (1994) document a reverse favorite-longshot bias in the moneyline market and conclude that the deviations are too small to bet on profitably. This literature is overwhelmingly pregame. The in-play evidence is thinner and mixed. Croxson and Reade (2014) find that soccer exchange prices update swiftly and essentially fully when goals arrive, consistent with semi-strong efficiency. More recent work documents imperfections in the price process itself: Simon (2024) rejects weak-form efficiency for MLB moneyline movement, finding systematic overreaction; Simon (2025) extends the negative autocorrelation of price changes to NFL, NBA, and NHL markets; and Angelini and De Angelis (2026) measure an approximately 0.64-for-one contemporaneous underreaction to benchmark probability changes in real-time prediction markets, with predictable subsequent drift.

Baseball performance. The times-through-order penalty entered the sabermetric canon through Tango, Lichtman, and Dolphin (2007). The strongest recent evidence favors continuous decay over a cliff: Brill, Deshpande, and Wyner (2023) find little support for a discontinuity at the third time through the order once batter and pitcher quality and other confounders are controlled, a conclusion our own analysis independently reproduces. Velocity effects on batting outcomes are real but small per mile per hour; Sutton-Brown (2023) estimates roughly 0.004 wOBA per mph operating through pitch quality, with a residual direct effect an order of magnitude smaller. Run values for discrete events, the RE24 table and linear weights, are standard tools from the same tradition.

Forecast evaluation. Our tests come from the econometric forecast evaluation literature. The pivotal one is the forecast encompassing framework of Chong and Hendry (1986): a benchmark forecast encompasses a rival when the rival adds no explanatory power for the target beyond the benchmark. Because our nested models are compared out of sample, we use the tools developed for that setting: the Diebold and Mariano (1995) comparison of predictive accuracy, West's (1996) asymptotic theory of predictive inference, the Clark and West (2007) correction for nested model comparison, and, for the conditional case, Giacomini and White (2006). Calibration diagnostics (reliability curves, the Brier score, expected calibration error) and the probabilistic interpretation of the area under the ROC curve (Hanley and McNeil, 1982) complete the toolkit. None of these instruments is new. Our contribution is to assemble them into one sequential protocol and to apply it with a sharp live betting line as the benchmark forecast.

The gap. These strands study the price series in isolation, or a single discrete shock, or baseball performance without reference to prices. To our knowledge, no published work combines live baseball state, pitch-level features, calibration analysis, an event-level transfer function, and a forecast encompassing test in which a sharp live betting line serves as the benchmark. That combination is what this paper adds.

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3. Methods

Our design compares two forecasts of the same quantity, the number of runs a game has left to score at a given moment, and asks whether publicly observable state improves the forecast already implied by the market. The unit of analysis throughout is the half-inning snapshot: a single moment in a single game at which both the market's live total and the full game state are observed. This section describes the data behind each snapshot (Section 3.1), the construction of the variables (Section 3.2), the escalating protocol by which a candidate is tested (Section 3.3), the estimators applied at each rung (Section 3.4), and the artifacts released for reproduction (Section 3.5).

3.1 Data

The study draws on 163 Major League Baseball games played in June 2026, each observed from three aligned sources. **Market prices.** Live full-game total (over/under) lines were recorded as one-minute trajectories from a single sharp-book feed. From each trajectory we take the main balanced total, the handicap at which the over and under prices sit closest to even. This is the market's median-outcome forecast of final total runs, and it serves as the incumbent forecast; its implied over probability feeds the calibration diagnostics. Appendix B details how a snapshot's line is matched to game state by timestamp. **Game state.** Complete play-by-play and boxscore records from the MLB Stats API supply, at every plate appearance, the inning and half, base-out state, score, batting order position, the identity and pitch count of the pitcher, and the number of times each batter has faced the current starter. **Pitch measurement.** Pitch-level release speeds from the same feed provide within-game velocity trajectories. Venue, weather (temperature, wind speed and direction relative to the field), and each game's realized final total complete the record. Sources are joined on game identifier and, for odds, on timestamp. The one-minute cadence and the single odds source are the principal constraints on what the design can measure; their consequences are collected in Section 6.

3.2 Variable construction

From the raw record we construct, at each snapshot, the two forecasts under comparison and the public state variables that might improve on them. The market forecast of remaining runs is B, the live total minus runs already scored. The realized remaining runs is Y, the final total minus runs already scored. The difference $Y - B$ is therefore the market's forecast error at that snapshot, and its predictability is the object of the study.

The candidate variables are built without reference to the outcome. The starter is the pitcher at the first plate appearance of each half; batting order slots are assigned by order of first appearance against that starter; times through the order counts prior batters faced, divided by nine. Velocity decline is computed two ways, across successive times through the order and within a fixed early window (the first twenty pitches versus the next twenty), a distinction that turns out to matter a great deal in Section 3.3. Bullpen quality is each team's season relief runs allowed per nine innings. Park factors and signed wind and temperature come from static published tables. The true change in run expectancy at each event, used by the transfer function, is ΔRE, runs scored plus the change in RE24 run expectancy across the play (Tango, Lichtman, and Dolphin, 2007). No variable uses information unavailable at the snapshot it describes.

3.3 The validation protocol

The core of the design is a fixed sequence of tests of increasing stringency, applied in the same order to every candidate. A variable is carried forward only until it is eliminated, and we report the rung at which elimination occurs. The rungs, in order: signal, robustness, out of sample, debiasing, conditional testing, forecast encompassing, transfer function.

The signal rung asks whether the variable is associated with scoring at all, in sample. The robustness rung asks whether that association survives reasonable changes in specification, thresholds, and subsample; an edge that moves with an arbitrary cutoff dies here. The out-of-sample rung re-estimates every model by leave-one-game-out cross-validation, so a variable is judged only on games not used to fit it. The debiasing rung replaces any post-treatment measurement, one defined only on a subsample selected by the outcome, with a pre-treatment analogue; a signal that survives in sample but vanishes when measured before treatment is diagnosed as selection, not effect. The conditional rung asks whether the variable earns its keep on average, not merely inside a hand-picked context such as hitter-friendly weather. The encompassing rung conditions on the market forecast itself and asks whether the variable adds anything to it. The transfer function rung, finally, asks not whether the market prices the variable but whether it prices it by the correct magnitude. One principle organizes the sequence: a betting hypothesis should be evaluated against the market, not merely against the outcome. The last two rungs are what enforce it. Figure 2 traces the study's actual path through the ladder.

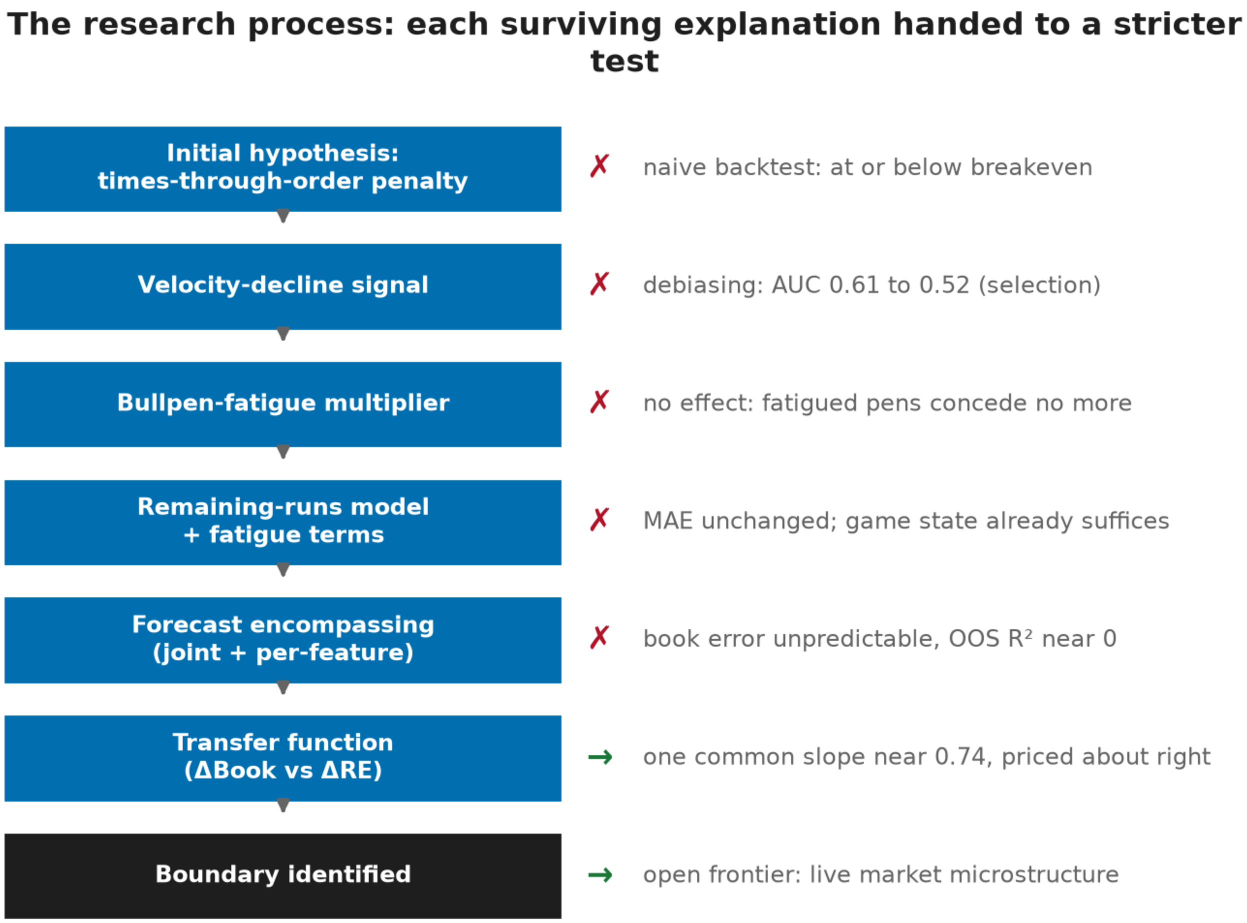

Figure 2. The research process. Each surviving explanation was handed to a stricter test. The sequence ends at a boundary, not at an edge.

3.4 Statistical evaluation

Each rung corresponds to a specific estimator, and everything is computed out of sample by leave-one-game-out.

Forecast encompassing (Chong and Hendry, 1986). For remaining runs Y, market forecast B, and a candidate feature set X, we fit three ridge-regularized linear forecasts, Y on B, Y on X, and Y on B and X together, and compare their out-of-sample R^2 and mean absolute error. If adding X to B does not improve on B alone, the market encompasses X. The sharpest form of the test regresses the market's forecast error $Y - B$ directly on X: if X cannot predict the error out of sample, it carries no information the price lacks. A per-feature variant fits Y on B plus each variable individually, so that two proxies for the same state cannot hide one another in the joint model. Continuous predictors are standardized; the ridge penalty is fixed in advance and applied to all non-intercept terms, and Appendix B confirms the conclusion is unchanged from ordinary least squares through heavy shrinkage. Because the restricted and unrestricted models are nested, a naive out-of-sample comparison of mean squared errors is biased against the larger model. We therefore report the Clark and West (2007) adjusted statistic, clustered by game, together with a 95 percent confidence interval for the encompassing gain obtained by block-bootstrapping whole games.

Why is encompassing the right test? Ordinary predictive accuracy cannot separate the two claims this paper must distinguish. A model that predicts runs well shows only that a variable is informative about the outcome; it says nothing about whether that information is already contained in an existing forecast. Encompassing answers the second question directly, by conditioning on the market forecast before evaluating the variable, so that what is being measured is incremental information rather than raw predictive power. A variable can predict the outcome while adding nothing beyond the market. As it happens, that describes every variable we tested.

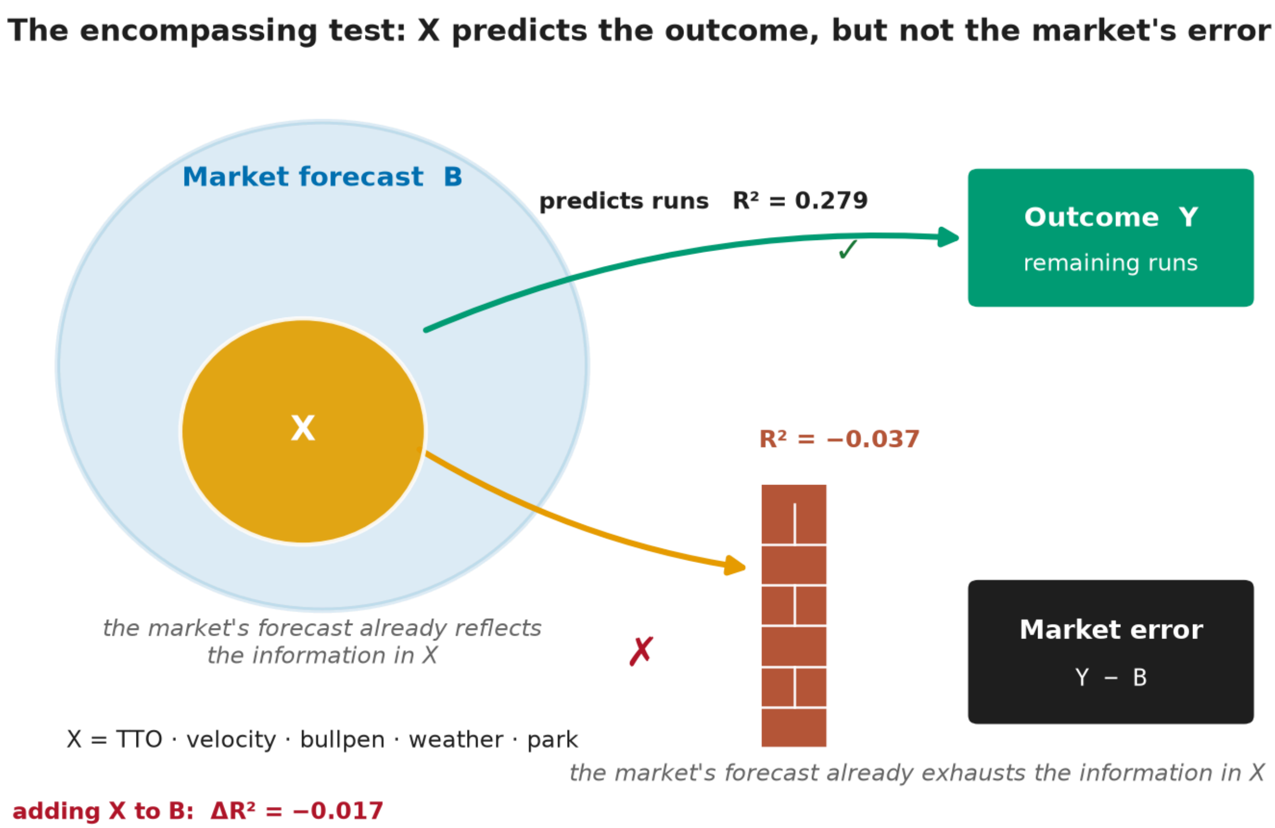

Figure 3. The forecast-encompassing test. The sharp market forecast B already reflects the information in the public features X (times through the order, velocity, bullpen, weather, park), drawn here as X contained within B. The features do predict the outcome Y, remaining runs, out of sample (R^2 of 0.279); but turned on the market's own forecast error, Y − B, they hit a wall (R^2 of −0.037), because the market's forecast already exhausts the information they carry. Adding X to B changes the out-of-sample R^2 by −0.017. Predicting the outcome, improving on the market, and turning a profit are three distinct questions, and every candidate in this study clears only the first.

Calibration. We assess the market forecast by binning snapshots on B and comparing mean realized Y in each bin against the identity line, and we assess probabilistic forecasts (for example, the probability that a team exceeds a run threshold) with reliability curves, the Brier score, expected calibration error, and the area under the ROC curve. The velocity debiasing of Section 3.3 is evaluated here as the change in out-of-sample AUC between a baseline forecast and forecasts augmented with the post-treatment and pre-treatment velocity measures; AUC confidence intervals use the analytic variance of Hanley and McNeil (1982).

Transfer function. For each in-game event we pair the true change in run expectancy ΔRE with the change in the live total one and five minutes later, and estimate the response ratio by event type together with a single common slope through the origin. Mean ΔRE by event type is checked against published linear weights as a validity control.

Statistical power. Because the paper's claims are null claims, we report what the design could have detected. For the ten-feature test of incremental information, 80 percent power at the 5 percent level corresponds to a minimum detectable incremental R^2 of roughly 0.007 if the 2,505 snapshots are treated as independent, rising to roughly 0.10 under the conservative assumption that only the 163 games are independent; the truth lies between. Every observed per-feature increment (at most 0.0018) sits below even the optimistic floor. In betting terms, detecting a 55 percent win rate against the 52.4 percent break-even at 80 percent power would require on the order of 2,000 wagers, so our few hundred qualifying situations can rule out large edges but not tiny ones. Appendix B gives the full calculation.

Uncertainty. All point forecasts are out of sample. Interval estimates use the Hanley and McNeil formula for AUC, Wilson intervals for proportions, and the bootstrap otherwise. Differences smaller than the width of their intervals are reported as such and are not interpreted as effects.

3.5 Reproducibility

Every quantity in this paper is recomputed from committed inputs by a fixed set of scripts, and estimation is deterministic, so results regenerate exactly. We release the cleaned data and feature schema as the Third Turn Benchmark Dataset (v1), together with reference implementations of the market forecast, the remaining-runs model, the encompassing tests, and the transfer function, an executable form of the protocol in Section 3.3. The frozen result files and the scripts that produce them are sufficient to reconstruct every figure and number without access to the original feeds.

The full pipeline, including the collector, is released at `[repository URL withheld for anonymous review]`, with citation metadata in `CITATION.cff`. Release commit `d709878` corresponds to the results reported here. The environment is Python 3.11.15 with dependencies pinned in `requirements-lock.txt`, and the analysis is versioned on three independent axes, Protocol 1.0, Collector 1.1, and Benchmark Dataset 2026.06, because method, engineering, and data evolve separately. From a clean checkout of that commit, installing the pinned requirements and running the scripts documented in `paper/REPRODUCE.md` regenerates every figure and the manuscript from the committed inputs.

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4. Results

The result is organized around a single test. We treat the sharp live line as the incumbent forecast and ask, following Chong and Hendry (1986), whether any public variable improves on it. Figure 4 answers that question directly; the remaining figures explain why the answer takes the form it does, and one of them, the velocity case in Figure 7, shows in miniature how a variable can predict the outcome and still add nothing to the price.

Figure 4 shows that publicly observable baseball variables provide no measurable incremental predictive information beyond the live market. Across 2,505 half-inning snapshots, a leave-one-game-out forecast of remaining runs from our full feature set achieves an out-of-sample R^2 of 0.279. The market-implied remaining total alone achieves 0.304. Combining the two changes R^2 by −0.017: adding the features to the market does not improve the forecast, and in fact slightly degrades it. The right panel repeats the test one feature at a time. The largest individual increment beyond the market is +0.0018 (bullpen quality); the rest sit near or below zero, with the largest negative values (park, temperature, wind) near −0.006, so no single variable is hiding behind the others. The sharpest version of the test regresses the book's forecast error, realized minus market-implied remaining runs, directly on the features. It is not predictable out of sample: R^2 of −0.037. This is the central empirical result of the study. The uncertainty around it is modest and quantified. The encompassing gain of −0.017 carries a 95 percent block-bootstrap confidence interval of [−0.036, +0.002], resampling whole games, and a Clark-West nested comparison does not reject equal predictive accuracy in the market's favor (statistic −0.1, one-sided p-value 0.55). If anything, the market alone is the better forecast. We emphasize that encompassing is the most demanding standard applied in this paper, because it conditions on the market's own forecast rather than on realized outcomes alone. A variable clears it only by improving a forecast that already reflects everything the market knows.

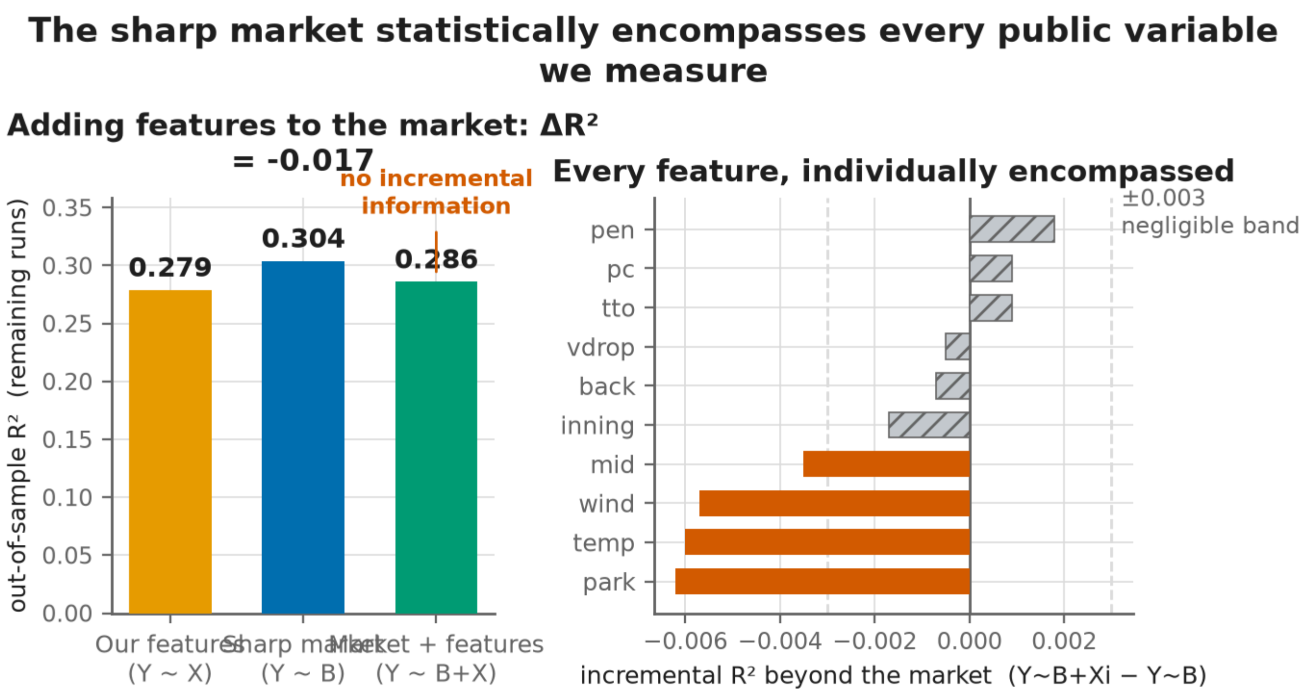

Figure 4. Forecast encompassing. Left: adding the feature set to the market forecast changes out-of-sample R^2 by −0.017. Right: each feature's individual incremental R^2 beyond the market; bars inside the 0.003 band are drawn in neutral gray.

Figure 5 shows that this boundary is a property of the whole class of public variables, not an accident of one unlucky choice. It arranges the candidate hypotheses from the public handicapping tradition (times through the order, velocity decline, bullpen fatigue multipliers, drop reversion in both directions, alternate-line skew, early-run anchoring, weather and park context, a remaining-runs fatigue term, and the joint encompassing test) against the successive tests each was put to: initial signal, robustness, out of sample, and the market test, with the verdict in the final column. What matters is not the tally but where the candidates fall. The fatal test varies by row, times through the order and the velocity signal at robustness, drop reversion and the joint model out of sample, the context variables at the market test, so that no single artifact, whether a coding error, one anomalous month, or one misspecified model, can be the common cause. The encompassing boundary is reached by many independent routes, which is the sense in which it is a property of the market rather than of any one variable. The full list, with each hypothesis's motivation and mode of elimination, appears in Appendix Table A1.

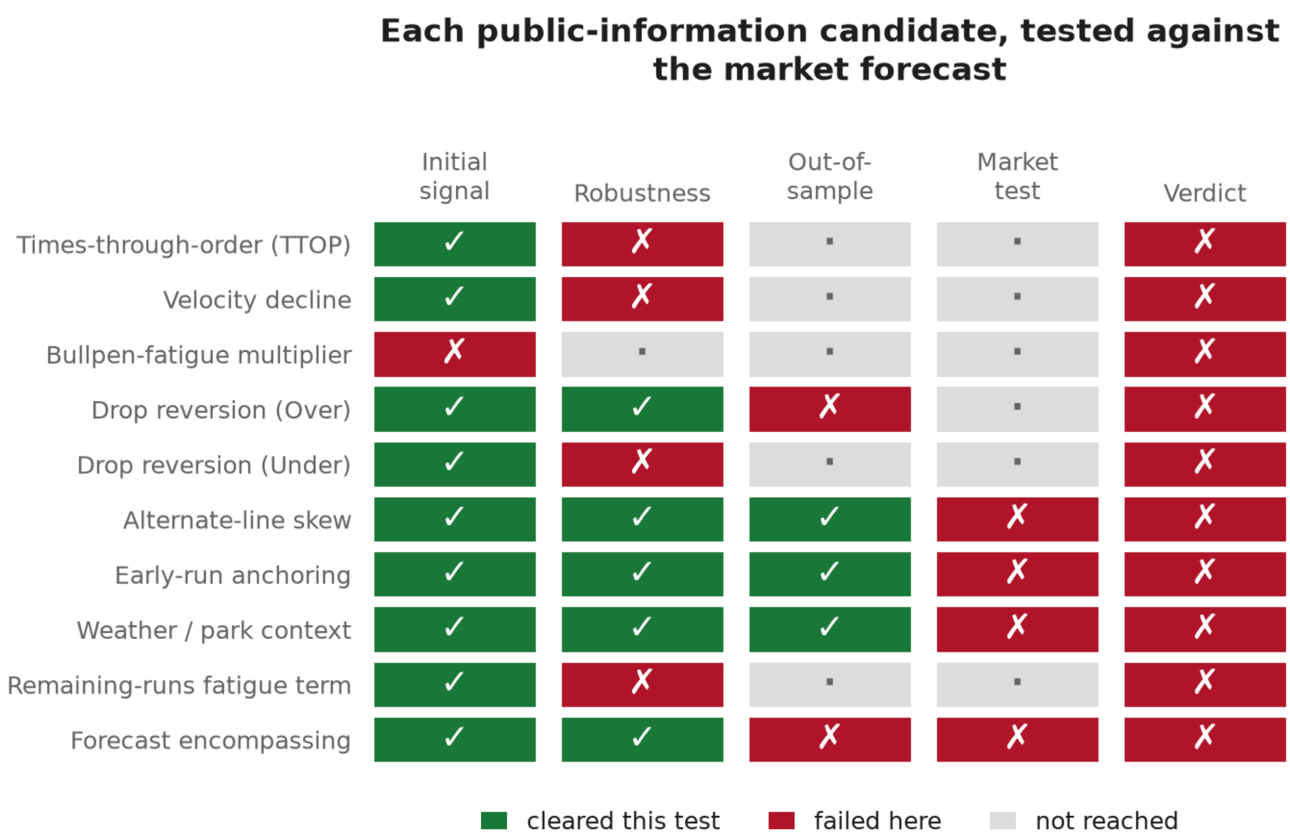

Figure 5. Where each public-information candidate fails relative to the market forecast, across successive tests. Green: cleared this test. Red: failed here. Gray: not reached.

Figure 6 restates the same evidence as a funnel, and isolates the one transition that carries the paper's economic content. Nine of the ten candidates produce a detectable in-sample association with runs, three survive out-of-sample validation, and none adds information beyond the market. The collapse at the final step, from variables that predict runs to variables that improve the price, is the distinction the encompassing test exists to draw: predicting runs is common, predicting a sharp market's error is the wall.[2] The counts derive directly from the Figure 5 classification, so the two figures cannot disagree.

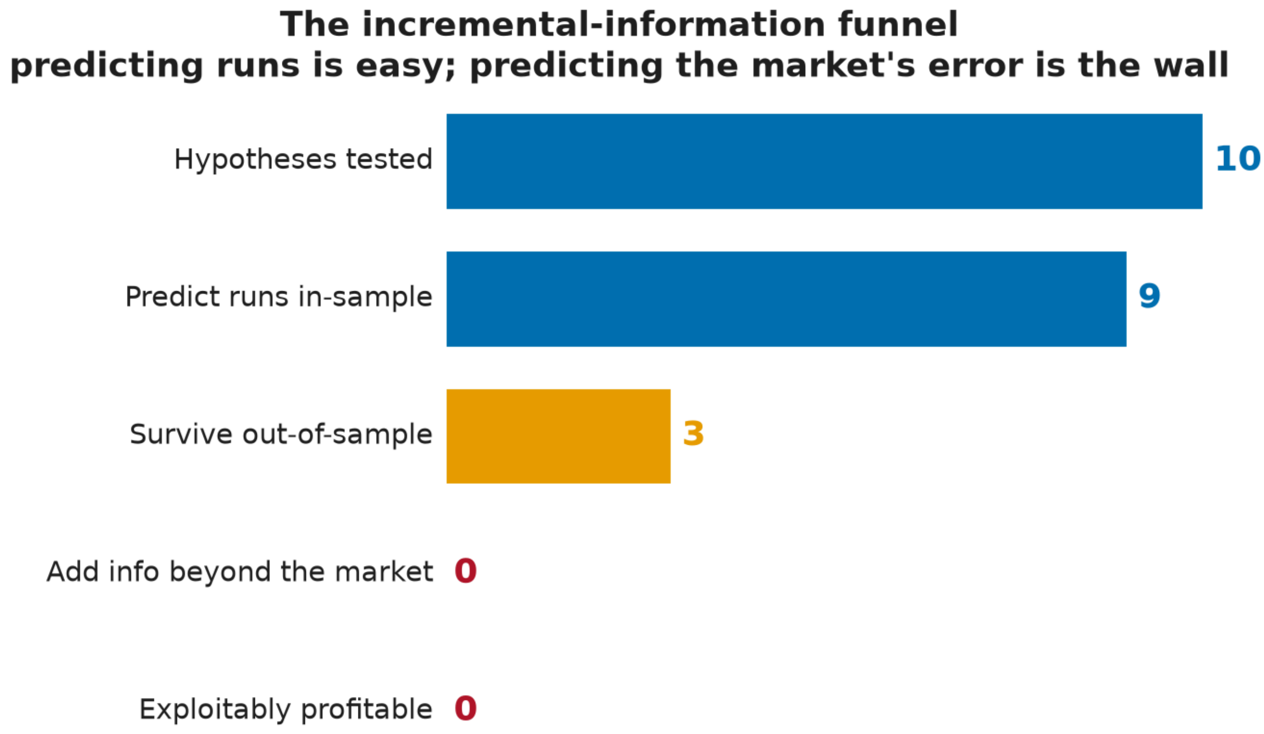

Figure 6. The incremental-information funnel: from predicting runs to improving the price. Counts derive from the Figure 5 classification.

Figure 7 develops the velocity case at length, because it is the clearest illustration in the study of how a variable can predict an outcome and still be worthless against the market, and because catching it is the single most important thing the protocol does. A model that adds the starter's velocity decline, measured from the first to the third time through the order, raises the out-of-sample AUC for the event "team scores more than 4.5 runs" from a baseline of 0.420 to 0.610. Taken at face value that is a large edge, and a study that stopped at out-of-sample prediction, as much betting research does, would report it as one. But the velocity-drop variable is post-treatment: it is defined only for starters who survived long enough to face the order a third time, which is to say, precisely the starters already being hit. Re-measuring velocity decline in a pre-treatment window (the first twenty pitches versus the next twenty) collapses the AUC to 0.524, with a confidence interval that straddles the 0.500 coin-flip line. The apparent signal was survivorship, not fatigue. This is exactly the class of artifact the debiasing rung exists to catch, and it is why the protocol places debiasing before the market test rather than after: a variable disqualified as a selection artifact never reaches the encompassing stage at all.

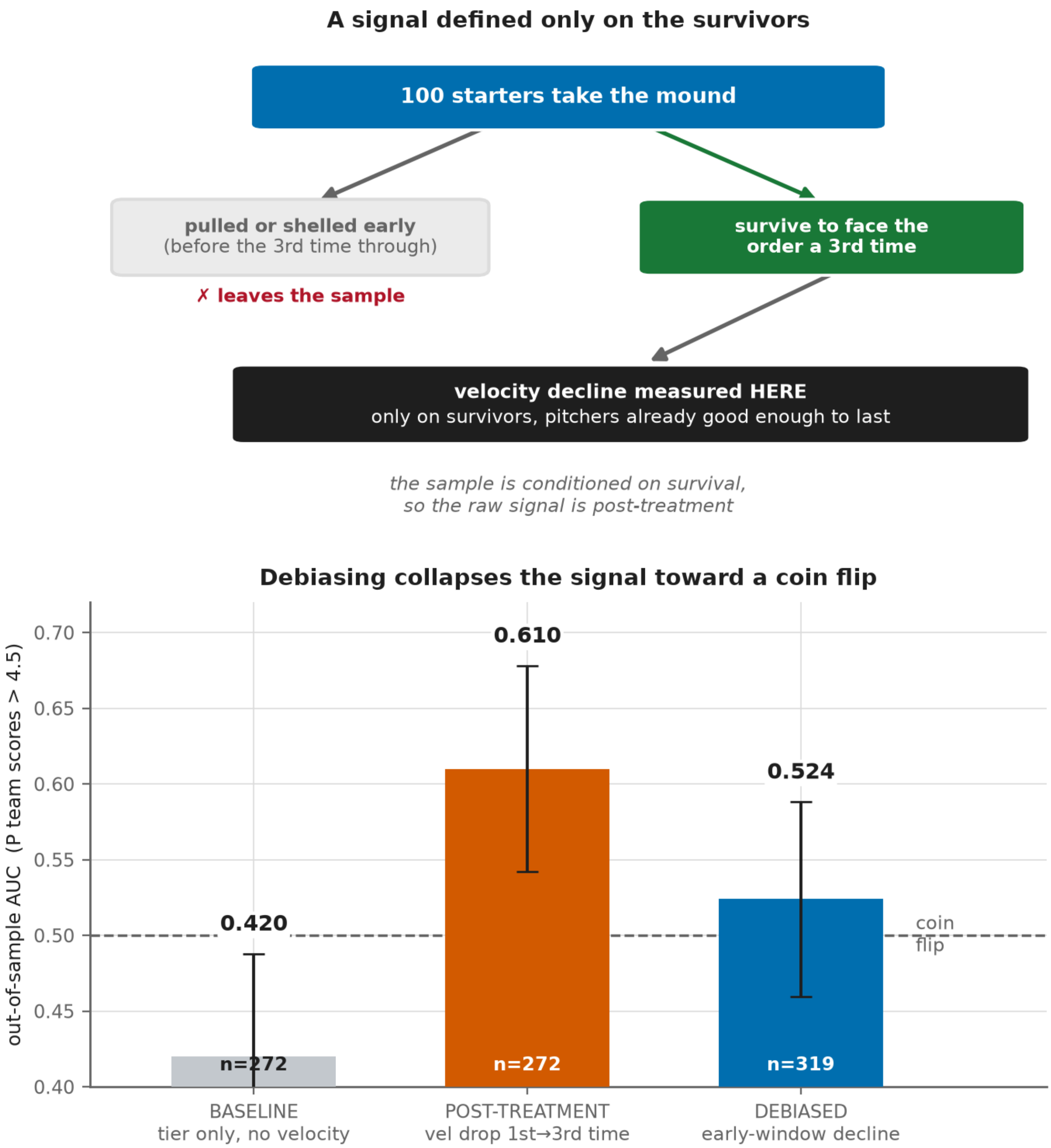

Figure 7. Post-treatment bias in the velocity signal. Left: the velocity-decline measure is defined only on starters who survive to face the order a third time, so the sample is conditioned on survival, precisely the pitchers already effective enough to last. Right: out-of-sample AUC for the event that a team scores more than 4.5 runs, with Hanley and McNeil 95 percent intervals. Conditioning on the third time through inflates the AUC to 0.610; measuring velocity decline instead in a pre-treatment early window collapses it to 0.524, an interval that straddles the coin flip.

Figure 8 turns from whether the market prices information to whether it prices it by the right amount. For each of 6,414 in-game events we compute the true change in run expectancy, ΔRE, and the change in the live total five minutes later. Plotted against each other, every positive-run event type lies on a single common slope of approximately 0.74 through the origin. The response ratios for the frequent events cluster in a narrow band from 0.63 to 0.84 (walk 0.63, double 0.64, single 0.72, home run 0.81, triple 0.84), with no ordering by event magnitude. That uniformity is evidence about the measurement pipeline rather than the market: a single source sampled at one-minute cadence acts as a low-pass filter that attenuates every shock by roughly the same factor. A per-event mispricing would show up as asymmetry across event types, and there is none. (Hit by pitch, with 115 events, sits below the band; pitching changes barely move the line, but RE24 cannot value the incoming reliever, so we exclude them from the elasticity claim.)

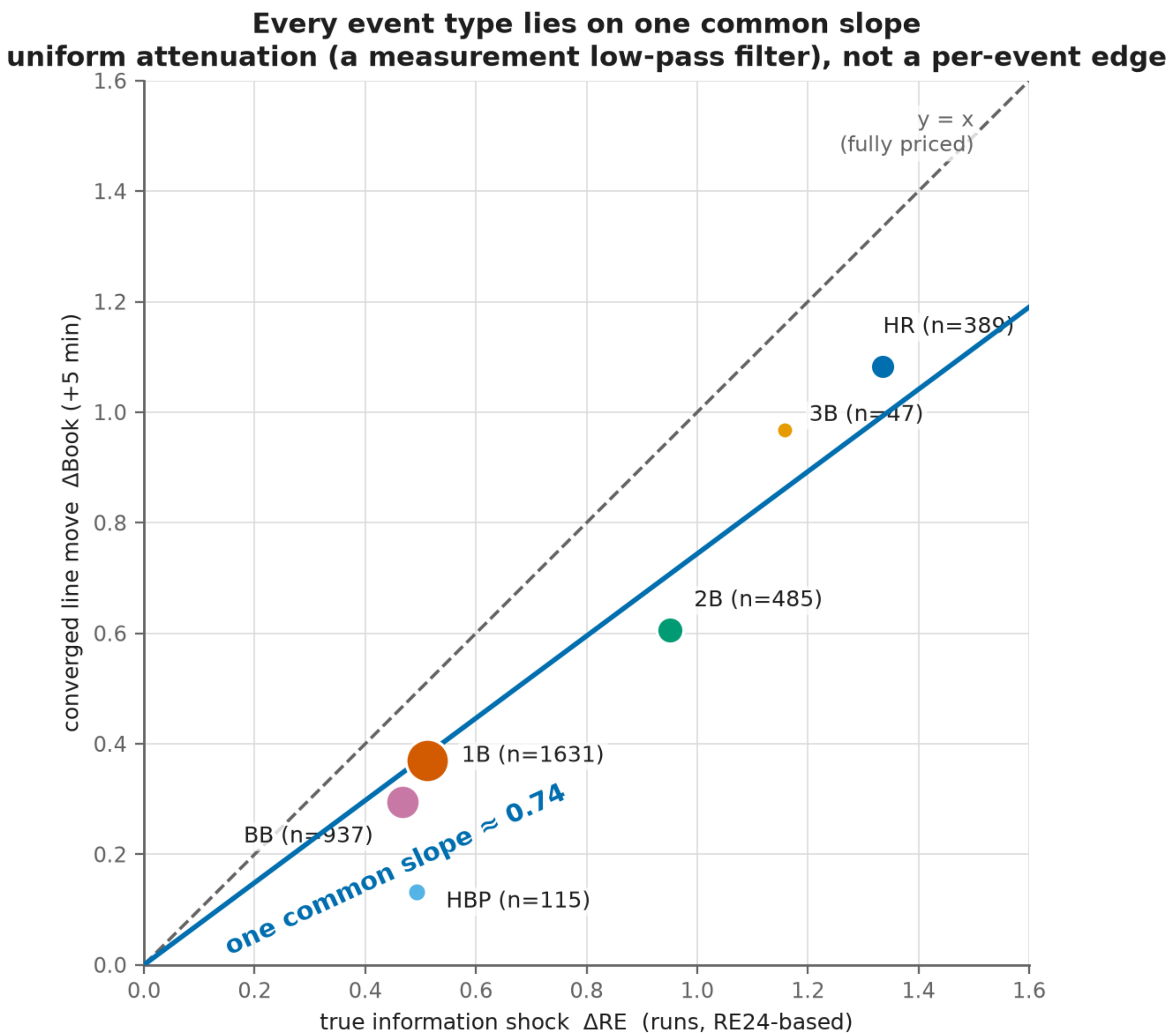

Figure 8. The market transfer function. Every positive-run event type lies on one common slope of roughly 0.74; marker area is proportional to event count.

Figure 9 closes the loop on the model side. The left panel bins the 2,505 snapshots by market-implied remaining runs and plots mean realized remaining runs in each bin. The points track the diagonal, so the market forecast is approximately calibrated within this sample. (The underlying leave-one-game-out remaining-runs model, fit on 2,859 snapshots, reaches an R^2 of 0.226, and adding fatigue terms leaves its mean absolute error essentially unchanged, about 0.001 runs, if anything slightly worse.) The right panel plots the distribution of the book's forecast error, and it contains the one number in this study that initially looked like a smoking gun. The error's median is zero, but its mean is +0.49 runs. The resolution is not market bias but arithmetic.[3] Remaining runs are right skewed (skewness +1.23), a balanced betting line sits at the median outcome, and realized runs and any least-squares forecast track the higher mean. The gap is a level term: regressing the error on the forecast gives a slope of +0.01, so any model with an intercept absorbs it, and it never enters the incremental information comparisons. Appendix B documents this in full. Beyond that intercept, the error is not predictable from any variable we measure. Together with Figure 4, this defines the empirical boundary of public-information prediction against a sharp live market, at the one-minute, single-book resolution of our data.

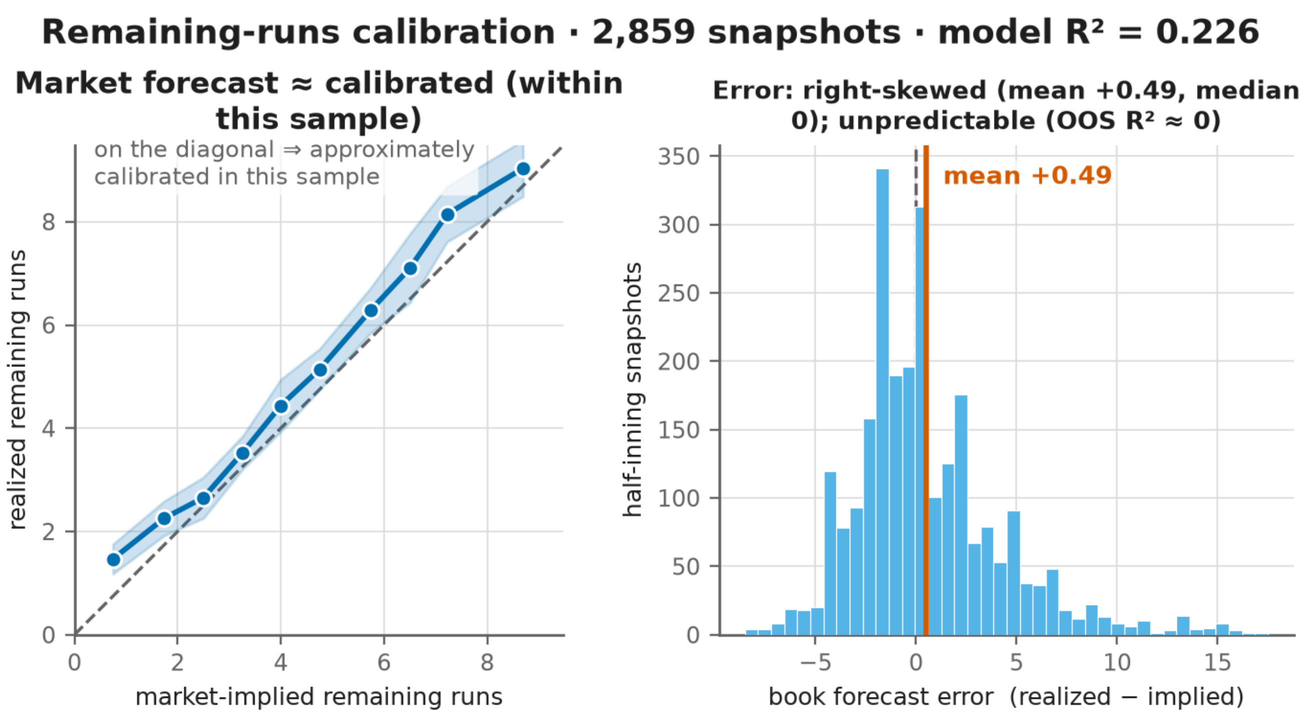

Figure 9. Market calibration. Left: binned market-implied versus realized remaining runs. Right: the distribution of the market's forecast error.

Summary. At every stage of the analysis, variables that predicted runs failed to provide incremental information once conditioned on the live market. The remaining sections interpret the boundary, draw out its methodological implications, and state what remains genuinely open.

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5. Discussion

The question was whether publicly observable baseball state variables contain incremental predictive information about remaining runs beyond the forecast embedded in a sharp live betting market. Within the limits of our data, they do not. This section says what that means, why it happens, why it matters outside baseball, and where the evidence runs out.

5.1 What the boundary means

The question was never whether baseball variables predict runs. They do. Pitch count, inning, pitcher quality, weather, park, and velocity all contain information about future scoring, and our own feature-only forecast (out-of-sample R^2 of 0.279) confirms it. The question was narrower and economically more meaningful: after conditioning on the forecast already embedded in a sharp live market, do these variables provide additional predictive information? The answer, in our data, is no.

That distinction disarms the most natural objection in advance. A reader may protest that weather obviously matters, or that a tiring pitcher obviously concedes more runs. Both are true, and neither is the point. Encompassing does not deny that these variables carry signal; it asks whether the signal is already in the price. When the market's forecast error cannot be predicted from any of these variables out of sample (R^2 of −0.037), the parsimonious reading is that the information is already incorporated in the market forecast. The variables are informative about runs and redundant with the price. Prediction survives. Increment does not.

Why should this be so? Not because baseball theory is wrong, but because the forecast embedded in a sharp live market is produced by participants with strong incentives to price observable state quickly, and it already reflects these variables. The transfer function evidence is consistent with that account: the line moves proportionately to the true change in run expectancy after every event type, with no class systematically mispriced. A forecast that adjusts proportionately to information shocks is exactly the kind whose residual carries no recoverable public-information signal, and that is what we observe.

5.2 Prediction is not profit

Here the paper stops being about baseball. Prediction and profit are distinct statistical problems, and conflating them is among the most common errors in applied betting research.

Much of that literature assumes a single arrow: better prediction leads to better betting. Our results break the arrow into three links that must each hold on their own. A variable may predict an outcome; velocity decline is associated with more runs. It may nonetheless carry no incremental information once the market forecast is conditioned on, because the price already reflects it. And even a variable that did carry incremental information would not automatically be profitable, since profit further requires that the edge exceed transaction costs, survive the vig, and persist after the act of betting moves the line. Prediction, increment, and profit are three questions, not one.

Encompassing matters because it isolates the middle link, the one betting studies most often skip. A naive backtest speaks to prediction. A profit-and-loss simulation speaks to the third link, and is easily flattered by overfitting and stale lines. Encompassing asks the middle question directly: does this variable improve on what the price already reflects? Because the test addresses increment rather than any particular staking scheme, our negative answer does not depend on how one would have bet.[4]

5.3 The efficient frontier of public information

We give the boundary a name: the efficient frontier of public information, the point at which additional publicly observable variables stop improving prediction once the market forecast is conditioned on. It is the live-market analogue of the classical statement of semi-strong efficiency, that price already reflects all public information, made operational as a forecast comparison rather than an event study and measured at the frequency at which the information actually arrives. Inside the frontier lie the observable baseball variables, pitch count, starter quality, bullpen, park, weather, velocity, times through the order, all of which the market encompasses. Outside it lie the dimensions our data cannot reach: the timing of price formation, disagreement across books, and the evolution of the full implied distribution rather than its mean.

One caution bounds the claim. What we observe is limited by the resolution of our instrument, a single sharp book sampled once a minute, and not necessarily by the resolution of the market. A lag that resolves within the minute, or a disagreement visible only across books, would be invisible to us and would register, incorrectly, as efficiency. Our result is therefore that public baseball variables carry no incremental information at the one-minute, single-book resolution of our data. Finer-grained or cross-book measurement could move the boundary, and mapping that is the subject of the follow-on work sketched in Section 7.

Every hypothesis in this study was, in effect, an attempt to cross the frontier with public state variables, and none succeeded. We do not read this as ten separate disappointments. It is one coherent measurement of where the frontier sits for one sport, one month, and one class of information, and it tells the next researcher where not to dig.

5.4 The methodological contribution

Although the motivation was baseball, nothing in the method is baseball-specific. The components, cross-validation, selection-bias correction, forecast encompassing, nested forecast comparison, are individually standard. The contribution is their integration into a single escalating protocol that separates variables which predict outcomes from variables that carry information not already reflected in an existing forecast. Each rung strips away one class of illusion: overfitting, then selection, then confounding, then redundancy with the market. A hypothesis that survives to the top has been tested against progressively harder alternatives rather than a single easy one.

The reason a protocol like this matters is the location of the burden of proof. Sports betting research frequently ends at the discovery of an in-sample signal. This study treated each positive result as a hypothesis requiring progressively stronger attempts at falsification. The protocol therefore shifts the burden from demonstrating prediction to demonstrating incremental information beyond an existing market forecast, a higher and more economically meaningful standard, and one that no single backtest can meet. That shift is a philosophy of evidence, not merely a workflow.

The ladder transfers unchanged to any market with a sharp public forecast and observable state: NBA totals, NFL spreads, soccer in play, tennis, racing. A researcher in any of these settings can adopt the same sequence, report the rung at which each candidate is eliminated, and compare results across domains. We call the sequence the Third Turn Protocol and release its reference implementation with the accompanying data as the Third Turn Benchmark Dataset (v1). A citable protocol and a shared dataset may prove more durable than any single finding, because they let a field accumulate falsifications rather than scattered one-off backtests.

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6. Limitations

We state the conditions under which the conclusion holds, without editorializing. **Scope.** The study covers 163 games in a single month (June 2026) of one sport. The boundary is characterized precisely, but only under those conditions, and we claim no seasonal or cross-sport generality. This sample is deliberately frozen: independent live collection is ongoing, and any temporal replication on a later, non-overlapping month would be reported separately rather than pooled into these estimates, so that such a replication remains a genuine out-of-sample test of the present result. No such replication is reported here. **Odds source.** Line trajectories come from a single sharp-book feed sampled at roughly one-minute intervals, so we cannot separate genuine price-formation latency from feed cadence, and the uniform sub-one response ratio in the transfer function is consistent with either. A companion study instruments the delivery path of live sportsbook feeds directly and measures what that caveat costs: delivered quote objects carry substantial, book-specific staleness, and the interval between publication and observation is not identifiable from public endpoints (Author 2026). That result bears on how the transfer-function timing should be read; it concerns when information entered prices rather than whether prices contain it, and leaves the estimates reported here unchanged. **Single-book benchmark.** All encompassing tests run against one sharp book; we cannot test cross-book agreement or leadership. **Market coverage.** Retail live team totals were not exposed by our feeds and are untested, as are first-five-inning totals. **Ground truth.** The remaining-runs model and the RE24 transfer benchmark use published static run values rather than park- or season-specific re-estimates, and the pitching-change response is reported but excluded from the elasticity claim because RE24 cannot value the incoming reliever. **Estimation.** Out-of-sample figures are leave-one-game-out, and effective sample sizes for the rarer events (triples, 47) are small; the corresponding ratios should be read accordingly. None of these conditions is load-bearing for the central result, that the book's forecast error is unpredictable from every variable we measure, but each bounds how far it generalizes.

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7. Remaining Questions

Distinct from the limitations, this section lists what the evidence genuinely does not answer: questions open not because the experiment was narrow but because they require data the historical record cannot provide. Does information propagate across books with a measurable lag, and is a laggard ever worth trading? Does the market update the shape of the implied run distribution, its variance, skew, and tails, as accurately as it updates the mean, or is higher-moment miscalibration where a residual edge hides? What is the information half-life of a shock: how long does the line take to absorb a home run, a pitching change, an injury? Does the boundary we map for full-game totals move when the market isolates the starters, as first-five-inning totals do? Each is a live-data question.

None is answerable from one-minute snapshots of a single book, and each is the natural subject of the market microstructure study that our forward-collected, timestamped price and state streams are being assembled to support. One conjecture worth stating, though we do not test it here, is that thinner or more retail-driven markets, live team totals and first-five-inning totals among them, may absorb public information less completely than the heavily traded full-game total, if only because lower volume leaves weaker incentive to correct a misprice; whether that holds is itself one of the questions the follow-on data are meant to settle.

8. What We Learned

This project began as a search for an exploitable feature of baseball. It ended by identifying the empirical boundary at which publicly observable baseball information stops providing incremental predictive value against a sharp live market. That boundary is itself a result. It redirects future work away from the discovery of additional baseball variables and toward the study of how information propagates through live betting markets. The contribution of this paper is therefore not a betting strategy but a reproducible procedure for telling the difference: a way to distinguish variables that predict outcomes from variables that carry information not already reflected in an existing forecast. We went looking for an edge and found a boundary. Of the two, the boundary has proved the more useful.

Appendix A. The full list of candidate hypotheses

Referenced from Section 4; Figure 5 is the main-text representation.

Hypothesis	Motivation	Test	Outcome	Mode of elimination
Times through the order	Familiarity penalty on the third time through	Binary gate, then gradient model; leave-one-game-out	Encompassed	Decay is continuous, not a cliff; out-of-sample fires below break-even; market prices it
Velocity decline	Fatigue shows up as lost velocity	Debiased early window versus post-treatment	Artifact	Defined only for starters who lasted long enough to be measured (selection); clean signal AUC near 0.52
Bullpen fatigue	Tired bullpen concedes more after a starter exits early	Isolated to the bullpen's own innings	No effect	Tired bullpens concede the same or fewer runs; no multiplier
Drop reversion (over)	A line that has fallen far reverts upward	Threshold sweep, all games	Not out of sample	Reversion is right skewed (win big, lose small); the median finishes below the line
Drop reversion (under)	The line stays low after a slow start	Banded splits plus robustness gates	Not robust	The effective band moves with the snapshot inning; concentrated in the recent subsample
Alternate-line skew	Buy the fat upper tail at longer odds	Empirical win rate versus efficient-implied	Priced	Empirical falls short of implied at every half-run increment; the tail is priced fatter than it realizes

Hypothesis	Motivation	Test	Outcome	Mode of elimination
Early-run anchoring	The live total underreacts to a first-inning scoring burst	Post-first-inning over, split by cause	Priced	49 of 50 bursts were hit-driven, leaving no fluky-runs population; the market prices the change
Weather and park	Books underprice hitter-friendly conditions	Conditional split	Priced	Hitter-friendly overs hit less often (46 versus 50 percent): the market over-adjusts
Remaining-runs fatigue	Fatigue improves a game-state model	Incremental MAE, leave-one-game-out	No increment	Game state already contains the information; the MAE change is nil
Joint encompassing	Does anything beat the market?	Y on B and X; error on X; per-feature variant	Encompassed	The book's forecast error is not predictable from any feature out of sample

Appendix B. Construction, power, and the forecast-error level

Snapshot construction. The unit of analysis is the first plate appearance of each half-inning through the eighth for which a live line is available. Game state (inning, half, base-out state, score, batting slot, pitcher, pitch count, times through the order) is read from the play-by-play at that plate appearance, and the market line is matched by timestamp, using the most recent quote at or before the plate appearance's start. The incumbent forecast is B, the main total minus runs already scored; the target is Y, the final total minus runs already scored.

Sample sizes. The three analyses use overlapping but distinct samples, by construction. The encompassing and calibration analysis uses 2,505 half-inning snapshots (innings one through eight with a matched line). The remaining-runs baseline uses 2,859 half-inning states; it does not require a market line, so a few more states qualify. The transfer function uses 6,414 events rather than snapshots, every run-scoring play and pitching change with a usable before and after line, which is a different unit entirely.

The forecast-error level. The book error $Y - B$ has mean +0.489 and median 0.000, on snapshots whose remaining-runs distribution has skewness +1.23. This is the mean-median gap of a right-skewed count distribution. A balanced betting line sits at the median outcome, while realized runs and any least-squares forecast track the higher mean. Regressing the error on B gives a slope of +0.014, so the gap is a constant level rather than a function of the forecast; any model with an intercept absorbs it, and it does not enter the incremental information comparisons. It is a property of the forecast's target functional, not evidence of market bias.

Nested forecast comparison. Comparing the nested out-of-sample forecasts by raw mean squared error is biased toward the smaller model; the Clark and West (2007) adjustment corrects this. The market model has the lower MSPE, 13.79 against 14.14. The Clark-West statistic, clustered by game, is −0.1 (one-sided p-value 0.55), so we do not reject equal predictive accuracy in the market's favor. Block-bootstrapping whole games, the encompassing gain is −0.017 with a 95 percent interval of [−0.036, +0.002].

Power. For the ten-feature test of incremental information, 80 percent power at the 5 percent level requires an incremental R^2 of about 0.007 when the 2,505 snapshots are treated as independent, and about 0.10 when only the 163 games are. Every observed per-feature increment (at most 0.0018) is below the former, feature by feature

(Figure B1). A 55 percent win rate against the 52.4 percent break-even would need roughly 2,000 wagers to detect at 80 percent power. The design excludes moderate incremental information, not arbitrarily small amounts.

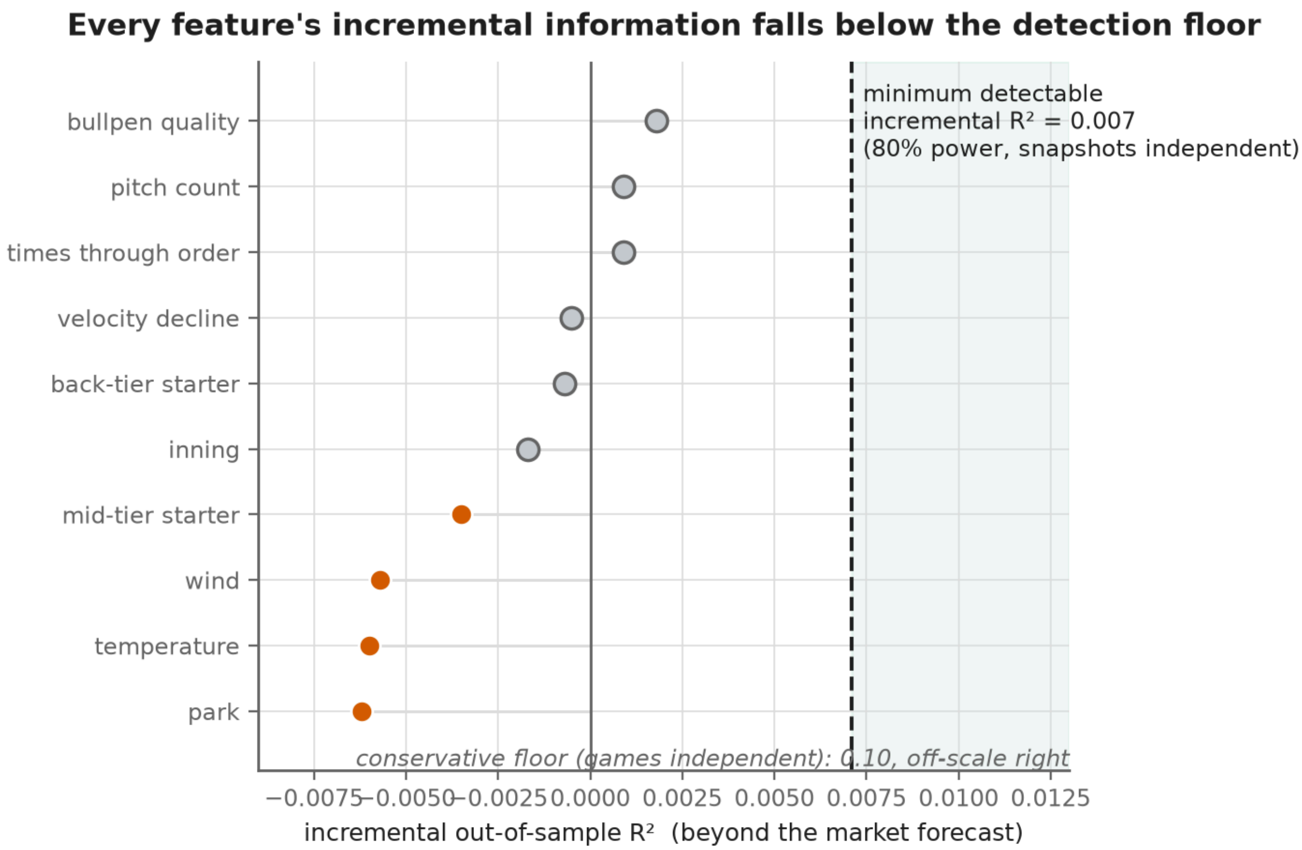

Figure B1. Every feature's incremental out-of-sample R^2 beyond the market forecast, against the minimum detectable effect. The largest increment is +0.0018 (bullpen); all fall well short of the 0.007 detection floor at 80 percent power when snapshots are treated as independent, and further still from the conservative floor of about 0.10 when only games are. The design could have detected a moderate incremental signal; none is present.

Penalty sensitivity. The encompassing conclusion is invariant to the ridge penalty. From ordinary least squares through heavy shrinkage, the encompassing gain stays between −0.017 and −0.013, and the out-of-sample R^2 for predicting the book error stays between −0.037 and −0.034. The features never improve on the market.

Ridge penalty	R^2, market	R^2, features	R^2, both	Gain	Error R^2
0 (OLS)	0.304	0.279	0.286	−0.017	−0.037
1	0.304	0.279	0.286	−0.017	−0.037
10	0.304	0.279	0.287	−0.017	−0.037
100	0.303	0.281	0.290	−0.013	−0.034

All Appendix B numbers are produced by a single committed script from the frozen inputs.

Appendix C. Transaction costs and the magnitude of the efficiency

Statistical detectability and economic profitability set different bars, and it is worth stating where the second one sits. Live totals carry a bookmaker's margin. At the standard price of −110 on each side, the bettor's break-even win rate is

BR = 110 / (110 + 100) ≈ 52.38 percent,

and the two prices imply an overround of roughly 4.8 percent; live in-play juice is often wider, so 52.38 percent is a floor on the hurdle rather than the hurdle itself. Profitability therefore requires not merely a positive edge but one large enough to move the win rate about 2.4 percentage points above a coin flip.

The information we measure does not approach even the weaker, statistical bar, let alone the economic one. The largest per-feature incremental out-of-sample R^2 beyond the market is 0.0018 (bullpen), and the market's forecast error is not predictable at all (R^2 of −0.037). Against the minimum detectable incremental R^2 of 0.007, under the optimistic assumption that snapshots are independent, every observed increment is smaller by a factor of several; under the conservative game-clustered assumption the floor is about 0.10, larger still. A detectable edge is a necessary but not a sufficient condition for profit, since profit further requires clearing the vig, so the economic hurdle is at least as demanding as the statistical one, and nothing clears even the statistical one. No observed variable comes within reach of a profitable edge. Figure C1 places the coin flip, the break-even hurdle, and a marginal winning edge on one scale.

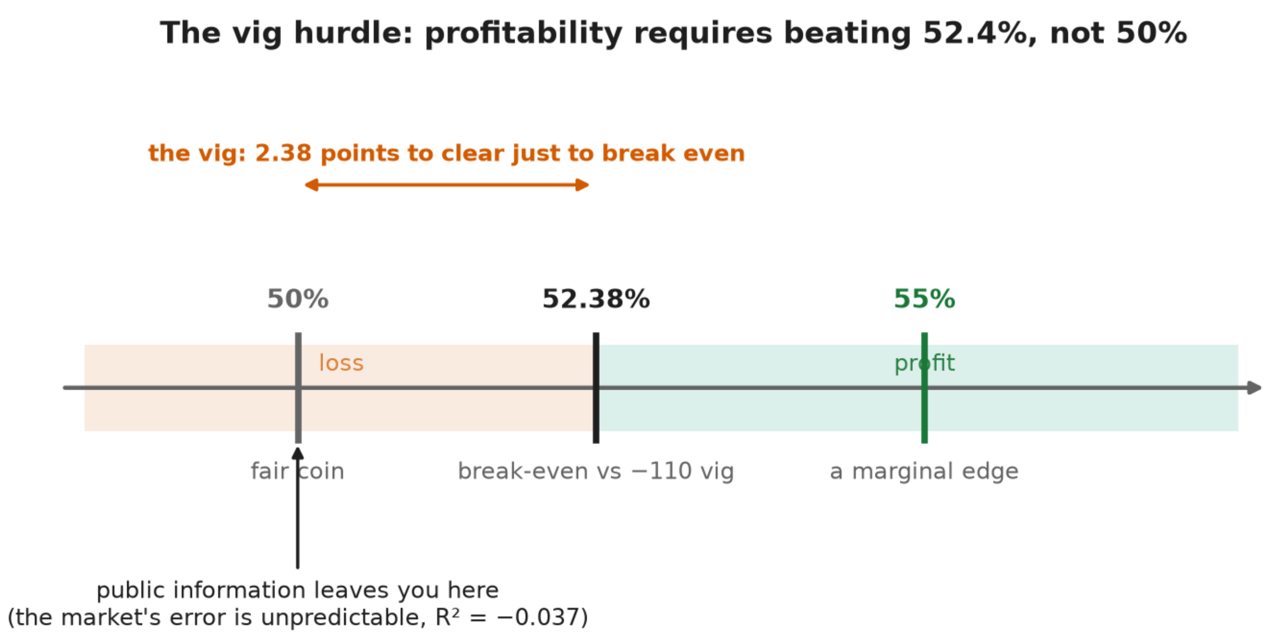

Figure C1. The vig hurdle in win-rate terms. A bettor must beat not the 50 percent coin flip but the 52.38 percent break-even implied by −110 pricing; the 2.38-point gap is the bookmaker's margin. The public information tested in this paper leaves the bettor at the coin flip, since the market's forecast error is unpredictable.

Data and code availability

The cleaned data, feature schema, and frozen result files are released as the Third Turn Benchmark Dataset (v1), openly available at `[repository URL withheld for anonymous review]` at release commit `d709878`, together with the Third Turn Protocol, its safeguard registry, and its objective stopping rules. Reference implementations reproduce every number reported here from the committed inputs. The third-party terms governing the underlying odds feed are under review; a persistent archival deposit and DOI will follow only if and when that review permits one.

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1. We came to the question honestly. The project began as an attempt to trade the times-through-order penalty, and the data infrastructure behind this paper was originally built to generate betting alerts. The name Third Turn refers to that third time through the batting order. Readers curious how the trading project became an efficiency paper may consult Figure 2, which is also a short autobiography.
2. Predicting runs is not hard. Predicting runs better than a firm that prices them for a living, and doing so after the vig, is a different occupation.
3. A forecast can look biased simply because the researcher asked it for the mean when it was offering the median. We spent an uncomfortable afternoon on this point.
4. We placed no wagers on the strength of these results. Given the results, this required no discipline whatsoever.